

SECTION 16351-A MEDIUM VOLTAGE GAS INSULATED SWITCHGEAR

PART 1 - GENERAL

1.1 SCOPE

A. The Contractor shall furnish and install the equipment as specified herein and as shown on the contract drawings.

1.2 REFERENCES

The metal-enclosed gas insulated switchgear and all components shall be designed, manufactured and tested in accordance with the latest applicable IEC standards of the following:

A. **Gas-insulated metal-enclosed switchgear.**

1. IEC 62271-1 High-voltage switchgear and controlgear common requirements
2. IEC 62271-200 High-voltage switchgear and control gear
3. IEC 61869-1 Instrument transformers - Part 1: general requirements
4. IEC 61869-2 Instrument transformers - Part 2: Additional requirements for current transformers
5. IEC 61869-3 Instrument transformers- Part 3: Additional requirements for inductive voltage transformers

B. **Circuit Breaker**

1. IEC 62271-100 High-voltage switchgear and controlgear - Alternating-current circuit breakers.

C. **Three-position switch for disconnect, isolation and grounding (earth).**

1. IEC 62271-102 High-voltage switchgear and controlgear - High-voltage alternating current disconnectors and earthing switches.

D. **Voltage Detecting System**

1. IEC 61243-5 Live working voltage detectors - voltage detecting systems.

1.3 SUBMITTALS - FOR REVIEW/APPROVAL

- A. The following information shall be submitted to the Engineer:
 - 1. Master drawing index.
 - 2. Front view elevation.
 - 3. Floor plan.
 - 4. Top view.
 - 5. Single line diagram.
 - 6. Nameplate schedule.
 - 7. Component list.
 - 8. Conduit entry/exit locations.
 - 9. Assembly ratings including:
 - a. Short-circuit rating.
 - b. Voltage.
 - c. Continuous current.
 - d. Basic impulse level for equipment over 600 volts.
 - 10. Major component ratings including:
 - a. Voltage.
 - b. Continuous current.
 - c. Interrupting ratings.
 - 11. Cable termination details.
 - 12. Product data sheets.
- B. Where applicable, the following additional information shall be submitted to the Engineer:
 - 1. Interlock scheme drawing and sequence of operations.
 - 2. Descriptive bulletins.

1.4 SUBMITTALS - FOR CONSTRUCTION

- A. The following information shall be submitted for record purposes:
 - 1. Final as-built drawings and information for items listed in Paragraph 1.03, and shall incorporate all changes made during the manufacturing process.
 - 2. Wiring diagrams.
 - 3. Certified production test reports.
 - 4. Installation information including equipment anchorage provisions.
 - 5. Seismic certification as specified.

1.5 DELIVERY, STORAGE AND HANDLING

- A. Equipment shall be handled and stored in accordance with

- manufacturer's instructions. One (1) copy of these instructions shall be included with the equipment at time of shipment.
- B. Shipping groups shall be designed to be shipped by truck, rail, or ship.
 - C. Switchgear shall be equipped to be handled by crane. Where cranes are not available, switchgear shall be suitable for forklifting using wooden shipping base or skidding in place on rollers using jacks to raise and lower the groups.
 - D. Switchgear being stored prior to installation shall be stored so as to maintain the equipment in a clean and dry condition. If stored outdoors, indoor gear shall be covered and heated.

1.6 OPERATION AND MAINTENANCE MANUALS

- A. Equipment operation and maintenance manuals shall be provided with each assembly shipped, and shall include instruction leaflets and instruction bulletins for the complete assembly and each major component.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Siemens.
- B. ABB.

The listing of specific manufacturers above does not imply acceptance of their products that do not meet the specified ratings, features and functions. Manufacturers listed above are not relieved from meeting these specifications in their entirety. Products in compliance with the specification and manufactured by others not named will be considered only if pre-approved by the Engineer ten (10) days prior to bid date.

2.2 RATINGS

The switchgear described in this specification shall be designed for operation on a 22.8 kV, three-Phase, three wire, low impedance grounded 60-hertz system. The switchgear assembly shall be rated for:

- A. Rated Voltage 24 kV rms
- B. Lightning Impulse Withstand Voltage 125 kV peak
- C. Power Frequency Withstand Voltage 50 kV rms, 60 Hz, 1-min
- D. Rated normal current for the busbar 2500 A rms

Note to Spec. Writer: Fill in data under 2.02B from Table 16351-A				
Table 16351-A				
Rated Maximum Voltage	Lightning Impulse Withstand Voltage	Rated Short Circuit Current at Rated Maximum Voltage (Short circuit breaking current)	3-Second Short Time Current Carrying Capability	Closing and Latching Capability (Momentary) (Peak withstand Current)
V kV rms	kV peak	I kA rms sym	kA rms sym	kA peak
4.76	60	25	25	65
4.76	60	31.5	31.5	82
8.25	95	25	25	65
8.25	95	25	31.5	82
15	95	25	25	65
15	95	31.5	31.5	82

27	125	25	25	65
27	125	31.5	31.5	82
38	170	25	25	65
38	170	31.5	31.5	82

- E. Circuit breakers are designated as Class S1 (circuit breaker intended to be used in a cable system). Each circuit breaker shall have the following ratings:

Rated Voltage: 24 kV

Continuous Current (Normal Current):
600, 1200, 1600, 2000, 3150 Ampere as shown on the drawings.

Short-Circuit Current (Short-circuit breaking current):
25 kA rms sym

Closing & Latching Capability (Peak withstand current):
65 kA peak

Short-time rating: 25 kA rms sym (3 sec)

Rated interrupting time is 5 cycles.

Out-of-Phase Breaking current is 6.25 kA.

Cable-Charging current is 31.5 A.

Electrical endurance is class E2 (extended electrical endurance), intended for use without auto-reclosing duty.

Mechanical endurance is class M2 (10000 mechanical operations).

Note: The circuit breaker is not rated to perform capacitor bank switching and has no assigned ratings.

2.3 CONSTRUCTION

- A. The switchgear assembly shall consist of individual vertical sections. Each vertical section shall contain a fixed sealed stainless steel tank filled with SF6 that contains the circuit breaker, three position (closed/opened/earthed) switch, and VT isolation switch as required for the application. Metal side sheets shall provide grounded barriers between adjacent structures. Switchgear vertical structures shall be one-high design (no more than one circuit breaker per vertical structure).

- B. Each vertical structure shall have a front accessible low voltage compartment. The low voltage compartment shall contain the gas pressure gauge, circuit breaker operating mechanism, three position switch operating mechanism and *[view port][camera] used for visual confirmation of the three position switch (closed/opened/earthed) status. The low voltage compartment shall also contain any meters, relays or other low voltage control devices required for the equipment. Additional control equipment that exceeds the available space on the low voltage compartment door shall be installed in an additional vertical section designated for low voltage devices

2.4 BUS

- A. The main bus shall be of the shielded solid insulation type and be copper, insulated with a silicone rubber and containing a metallic ground screen. The bus shall be a modular design in order to support field installation using factory installed mounting locations and hardware. Bus shall be mounted on the top of the switchgear in a continuous run. The switchgear shall be constructed so that all buses, bus supports and connections shall withstand stresses that would be produced by currents equal to the momentary ratings (peak withstand) of the circuit breakers. Main bus shall be rated as shown on the drawings. The main bus shall have provisions for future extension. The temperature rise of the bus and connections shall be in accordance with IEC standards and documented by design tests.
- B. A copper ground bus shall extend the entire length of the switchgear.

2.5 WIRING/TERMINATIONS

- A. The switchgear manufacturer shall provide suitable terminal blocks for secondary wire terminations, with a minimum of 10% spare terminals. One control circuit cutout device shall be provided for each circuit breaker. Switchgear secondary wire shall be #14 AWG, type SIS rated 600 volt, 90 degrees C, furnished with wire markers at each termination. Wires shall terminate on terminal blocks with marker strips numbered in agreement with detailed connection diagrams.
- B. Incoming line and feeder bolted tee connections of the type and size indicated elsewhere shall be furnished by contractor.

2.6 CIRCUIT BREAKERS

- A. The circuit breaker primary current path shall be fixed within the gas insulated stainless steel tank. The circuit breaker operating mechanism shall be outside of the gas tank and accessible from the front. The circuit breaker shall be operated by a motor-charged stored energy spring mechanism, charged normally by a universal electric motor and in an emergency by a manual handle.
- B. Each circuit breaker shall contain three vacuum interrupters separately mounted in a self-contained pole unit. The current transfer from the vacuum interrupter moving stem to the circuit breaker main conductor shall be a non-sliding design. The circuit breaker front panel shall be removable for ease of inspection and maintenance.
- C. The secondary contacts shall be terminated to terminal strips for interconnection with the assembly.
- D. Interlocks shall be provided to prevent closing of the circuit breaker while the three position switch is being operated manually. The three position switch cannot be operated unless the circuit breaker is in the open position.
- E. The circuit breakers shall be electrically operated by the following control voltages:

125 volt DC close.

For each circuit breaker, provide green and red LED lights to indicate circuit breaker open and closed status.
- F. DC control voltage shall be supplied by external source.
- G. Leakage of gas from the circuit breaker tank shall not exceed 0.1% of the gas per compartment per year.

2.7 THREE POSITION SWITCH

- A. The three position switch shall be fixed within the gas insulated stainless steel tank. The switch mechanism shall be outside of the gas tank and accessible from the front. The switch shall be operated by a motor operator mechanism and provided with interlocking to the circuit breaker. The switch shall be capable of being operated manually.
- B. The three position switch shall have three distinct positions of closed, opened, and ground (earthed). The switch shall have a visible indicator to provide positive indication of its status.

- C. The three position switch shall have auxiliary contact arrangements for switch status, electrical interlocking, and indicating lights. Spare contacts shall be available for customer or future use.
- D. The three position switch shall be electrically operated by the following control voltages:

125 volt DC operating motor.

For each three position switch, provide green, red, and amber LED lights to indicate switch open, closed, and earthed (grounded) status.

2.8 AUXILIARY DEVICES

- A. Ring type current transformers shall be furnished as indicated on the drawings. The thermal and mechanical ratings of the current transformers shall be coordinated with the circuit breakers. The standard location for the current transformers is on the bus side and cable side of the units. Cable side CTs shall be front accessible to permit adding or changing current transformers. Bus side CTs shall be top accessible to permit adding or changing current transformers. Shorting terminal blocks shall be furnished on the secondary of all the current transformers.
- B. Metal-clad voltage transformers (VT) of the quantity and ratings indicated in the detailed specification shall be supplied. Voltage transformers can be top-mounted and fused, bottom-mounted and unfused, or remote mounted and either fused or unfused.
 - 1. Top-mounted voltage transformers shall be installed and contained in an enclosed top accessible auxiliary compartment to allow removal of the VT or fuse, if required. The SF6 gas tank shall contain a VT isolating switch to connect and ground (earth) top-mounted switchgear VTs from the primary circuit. Connection of the VTs shall be via insulated inner cone connectors.
- C. Where indicated on the drawings, provide surge protection. Surge protection, if required, shall be constructed with metal (zinc) oxide varistor technology housed in a bolted tee connector.

2.9 ENCLOSURES

- A. The switchgear described in these specifications shall be indoor construction, with devices arranged as shown on drawings.

2.10 NAMEPLATES

- A. An engraved ratings nameplate shall be furnished for each switchgear vertical section giving information in accordance with IEC 62271-200.
- B. Engraved circuit nameplates, mounted on the face of the assembly, shall be furnished for all main and feeder circuits as indicated on the drawings. Nameplates shall be laminated plastic, black characters on white background, and secured with screws. Characters shall be 3/16-inch high, minimum. Circuit nameplates shall be provided with circuit designations as shown on purchaser's single-line diagrams.
- C. Control components mounted within the assembly, such as fuse blocks, relays, pushbuttons, switches, etc., shall be suitably marked for identification corresponding to appropriate designations on manufacturer's wiring diagrams.

2.11 FINISH

- A. The finish shall consist of a coat of light gray textured (RAL-7035), thermosetting, polyester powder paint applied electrostatically to pre-cleaned and phosphatized steel and aluminum for internal and external parts.

2.12 ACCESSORIES

- A. The switchgear manufacturer shall furnish accessories for test, inspection, maintenance and operation, including: control fuses, indicating lights, spare relays.

2.13 PARTIAL DISCHARGE TESTS

- A. The switchgear shall be corona free. It shall be tested for partial discharges in accordance with IEC 62271-200. The corona discharges measured during the tests shall be less than 100 pico coulombs.

2.14 POWER QUALITY ANALYSIS INSTRUMENT

- A. There must be 2 power quality analysis instruments on the incoming side.

PART 3 - EXECUTION

3.1 FACTORY TESTING

- A. The following standard factory tests shall be performed on the circuit breaker provided under this section. All tests shall be in accordance with the latest version of IEC standards.
 - 1. Circuit breakers operated over the range of minimum to maximum control voltage.
 - 2. One-minute power frequency withstand test per IEC standards.
 - 3. Final inspection and quality check.
- B. The following production test shall be performed on each switchgear vertical section:
 - 1. One-minute power frequency withstand test per IEC standards on primary and secondary circuits.
 - 2. Operation of wiring, relays and other devices verified by an operational sequence test.
 - 3. Final inspection and quality check.
- C. The manufacturer shall provide three (3) certified copies of factory test reports.
- D. Factory tests as outlined above under 3.01.B shall be witnessed by the owner's representative.
 - 1. The manufacturer shall notify the owner two (2) weeks prior to the date the tests are to be performed.

3.2 FIELD QUALITY CONTROL

- A. Provide the services of a qualified factory-trained manufacturer's representative to assist the Contractor in installation and startup of the equipment specified under this section. The manufacturer's representative shall provide technical direction and assistance to the contractor in general assembly of the equipment, connections and adjustments, and testing of the assembly and components contained therein.
- B. The Contractor shall provide three (3) copies of the manufacturer's field startup report.

3.3 MANUFACTURER'S CERTIFICATION

- A. A qualified factory-trained manufacturer's representative shall certify in writing that the

equipment has been installed, adjusted and tested in accordance with the manufacturer's recommendations.

- B. The Contractor shall provide three (3) copies of the manufacturer's representative's certification.

3.4 TRAINING

- A. The Contractor shall provide a training session for up to five (5) owner's representatives for 5 normal workdays at a job site location determined by the owner.
- B. The training session shall be conducted by a manufacturer's qualified representative. Training program shall include instructions on the assembly, circuit breaker, protective devices, and other major components.

3.5 INSTALLATION

- A. The Contractor shall install all equipment per the manufacturer's recommendations and contract drawings.
- B. All necessary hardware to secure the assembly in place shall be provided by the Contractor.

3.6 FIELD TESTING

- A. The field testing shall include the following:
 - 1. Power frequency withstand test (at 80% of the prescribed power frequency withstand voltage).
 - 2. Insulation resistance.
 - 3. Functional test of the fully installed and wired equipment delivered.

- END -